

EEE4114F

Introduction to Reinforcement Learning

Reinforcement Learning

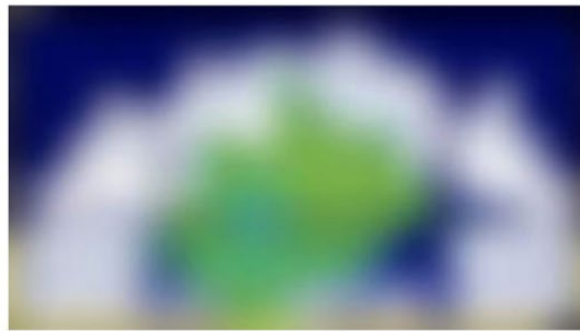
- Learn through interaction
 - Learn desirable behaviour over time
 - Can affect the environment
 - Different from supervised learning we've encountered
 - E.g. supervised learning is all based on pre-collected data. No opportunity to affect environment
- Goal-directed
 - Numerical reward signals
 - Learn actions that yield maximum reward

Examples in Engineering

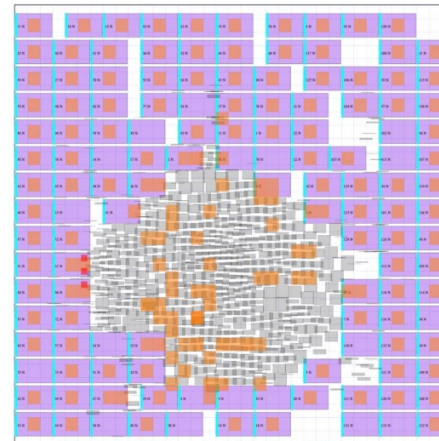
- Ibarz, J. et al. (2021) 'How to train your robot with deep reinforcement learning: lessons we have learned', The International Journal of Robotics Research, 40(4–5), pp. 698–721. doi: 10.1177/0278364920987859.
- Mirhoseini, A., Goldie, A., Yazgan, M. et al. A graph placement methodology for fast chip design. Nature 594, 207–212 (2021). <https://doi.org/10.1038/s41586-021-03544-w>



Extended Data Fig. 5 | Visualization of a real TPU chip. Human expert placements are shown on the left and results from our approach are shown on the right. The white area represents macros and the green area represents standard cells. The figures are intentionally blurred because the designs are

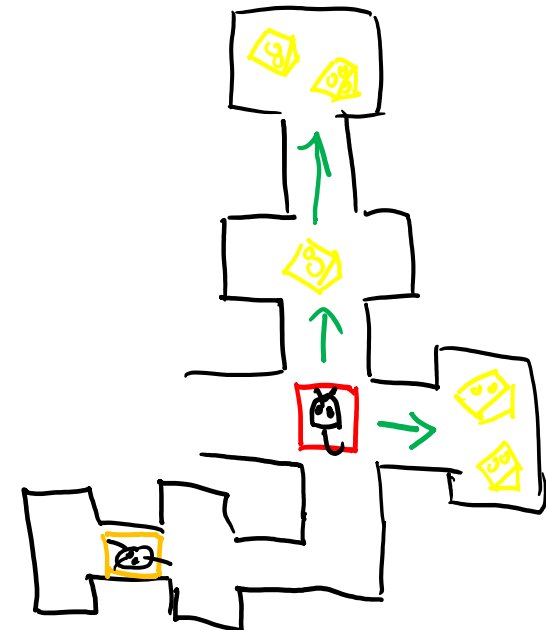


proprietary. The wirelength for the human expert design is 57.07 m, whereas ours is 55.42 m. Furthermore, our method achieves these results in 6 h, whereas the manual baseline took several weeks.



Elements of RL

- Policy
 - How does the agent behave in a given situation
- Reward signal
 - Numerical reward
 - Determined by environment (feedback)
 - Objective is to maximise reward
- Value function
 - Long-term desirability of a state or action in a state
 - Could have low immediate reward, but could be good long-term
- Model*
 - Do you know how the environment behaves beforehand?
 - Allows for planning



Next up:

Finite Markov Decision Processes